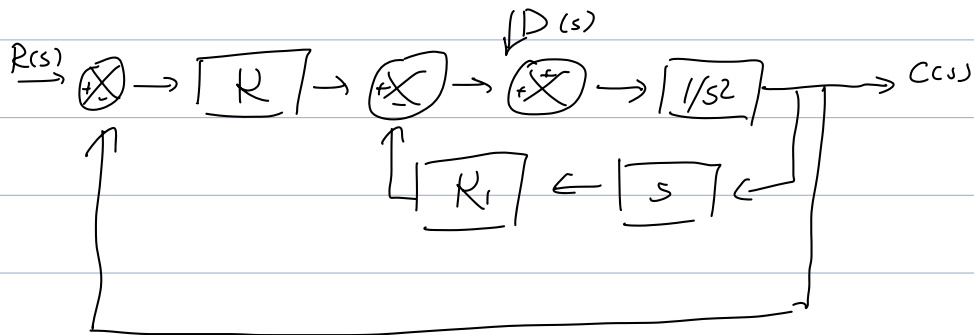
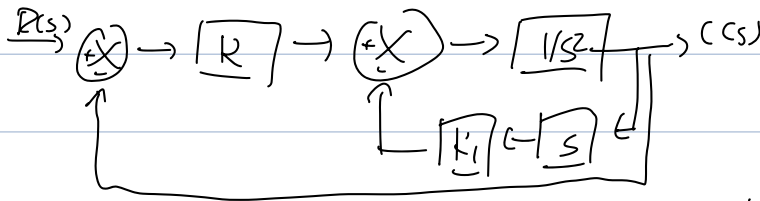


T_1

Q1.

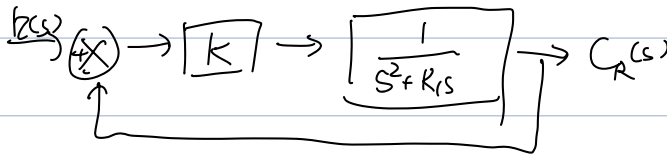


With $D(s) = 0$, we have:

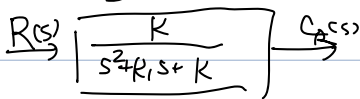


Eliminating the inner loop:

$$\frac{1/s^2}{1 + \frac{1}{s^2} K_1 s}$$



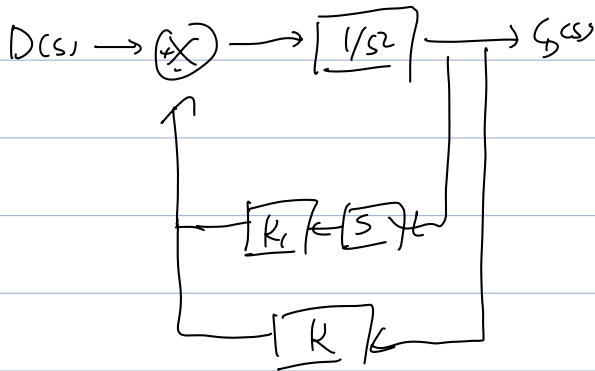
Eliminating final outer loop:



$$\left[\text{For outer loop: } \frac{\frac{K}{s^2 + K_1 s}}{1 + \frac{K}{s^2 + K_1 s}} = \frac{K}{s^2 + K_1 s + K} \right]$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{K}{s^2 + K_1 s + K}$$

With $R(s) = 0$, the system can be redrawn as shown:



Eliminating the loop:

$$\frac{1/s^2}{1 + (K/s + K) \cdot \frac{1}{s^2}} = \frac{1}{s^2 + Ks + K}$$

$$D(s) \rightarrow \left[\frac{1}{s^2 + Ks + K} \right] \rightarrow G(s)$$

$$\Rightarrow \frac{G(s)}{D(s)} = \frac{1}{s^2 + Ks + K}$$

$$\text{Therefore } C(s) = R(s) + G(s) = \frac{1}{s^2 + Ks + K} (KR(s) + D(s))$$

For a unit step: $R(s) = \frac{1}{s}$, $D(s) = 0$

$$\Rightarrow C(s) = \frac{100}{s(s^2 + 12s + 100)}$$

$$\begin{aligned} \frac{100}{s(s^2 + 12s + 100)} &= \frac{A}{s} + \frac{Bs + C}{s^2 + 12s + 100} = \frac{A(s^2 + 12s + 100) + (Bs + C)s}{s(s^2 + 12s + 100)} \\ &= \frac{(A+B)s^2 + (12A+C)s + 100A}{s(s^2 + 12s + 100)} \end{aligned}$$

$$\Rightarrow \begin{cases} A+B=0 \\ 12A+C=0 \\ 100A=100 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=-1 \\ C=-12 \end{cases}$$

$$\Rightarrow C(s) = \frac{1}{s} - \frac{s+12}{s^2+12s+100}$$

$$= \frac{1}{s} - \frac{s+6}{(s+6)^2+8^2} - \frac{6}{8} \cdot \frac{8}{(s+6)^2+8^2}$$

$$\therefore c(t) = 1 - e^{-6t} \cos 8t - \frac{3}{4} e^{-6t} \sin 8t$$

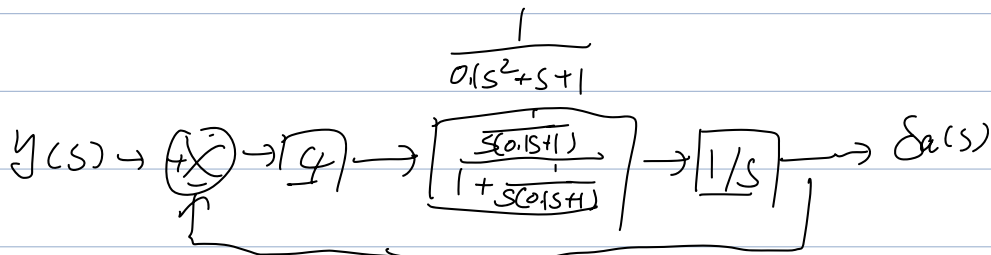
$$= 1 - e^{-6t} \left(\cos 8t + \frac{3}{4} \sin 8t \right)$$

For $\phi(s) = \frac{1}{s}$,

$$C(s) = \frac{1}{s(s^2+12s+100)}$$

$$\Rightarrow c(t) = 0.01 \left(1 - e^{-6t} \left(\cos 8t + \frac{3}{4} \sin 8t \right) \right)$$

Q2.



$$Y(s) \rightarrow \left[\frac{4}{1 + \frac{4}{s(0.1s^2+s+1)}} \right] \rightarrow S_a(s)$$

$$Y(s) \rightarrow \left[\frac{4}{0.1s^3 + s^2 + s + 4} \right] \rightarrow S_a(s)$$

$$\frac{Y(s)}{S_a(s)} = \frac{40}{s^3 + 10s^2 + 10s + 40}$$

Q4. (iii) Poles: $(s^2 + s - 6)(s^2 + 2s + 10) = 0$

$$\Leftrightarrow (s-2)(s+3) \left(s \pm \frac{-2 \pm \sqrt{4-40}}{2} \right) = 0$$

$$s \pm (-1 \pm 3i)$$

$$\Rightarrow s_1 = 2 \quad s_2 = -3 \quad s_3 = -1 + 3i \quad s_4 = -1 - 3i$$

not stable, oscillatory

zeros from:

$$(iv). -1, -4, 3 \pm \sqrt{9-25}$$

$$3 \pm 4i$$

not stable

Q6. i)

$$\begin{vmatrix} 1 & 1 \\ 4 & 0 \\ 1 & \end{vmatrix}$$

$$b_1 = -\frac{1}{4} \cdot \begin{vmatrix} 1 & 1 \\ 4 & 0 \end{vmatrix} = 1$$

no change of sign in 1st column

$\therefore$ system is stable

ii)

$$\begin{vmatrix} 1 & 5 \\ 4 & 6 \\ \frac{7}{2} & 0 \\ 6 & \end{vmatrix}$$

$$b_1 = -\frac{1}{4} \begin{vmatrix} 1 & 5 \\ 4 & 6 \end{vmatrix} = -\frac{1}{4}(6-20) = \frac{7}{2}$$

$$c_1 = -\frac{1}{7/2} \begin{vmatrix} 4 & 6 \\ 7/2 & 0 \end{vmatrix} = -\frac{2}{7}(-\frac{7}{2} \times 6) = 6$$

no sign changes, stable

iv)

$$\begin{vmatrix} 1 & 2 & 8 \\ 1 & 4 & 0 \\ -2 & 8 & \end{vmatrix}$$

$$b_1 = -\frac{1}{1} \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = -(4-2) = -2$$

$$b_2 = -\frac{1}{1} \begin{vmatrix} 1 & 8 \\ 1 & 0 \end{vmatrix} = -\frac{1}{1}(0-8) = 8$$

$$\text{Q7. i) } \begin{array}{l|lll} s^4 & 1 & 1 & K \\ s^3 & 3 & 2 & 0 \\ s^2 & 1/3 & K & 0 \\ s^1 & -9K+2 & 0 & \\ s^0 & K & & \end{array}$$

$$b_1 = -\frac{1}{3} \begin{vmatrix} 1 & 1 \\ 3 & 2 \end{vmatrix} = -\frac{1}{3} (2-3) = \frac{1}{3}$$

$$b_2 = -\frac{1}{3} \begin{vmatrix} 1 & K \\ 3 & 0 \end{vmatrix} = -\frac{1}{3} (-3K) = K$$

$$c_1 = -\frac{1}{1/3} \begin{vmatrix} 1/3 & 2 \\ 1/3 & K \end{vmatrix} = -3(3K - \frac{2}{3}) = -9K+2$$

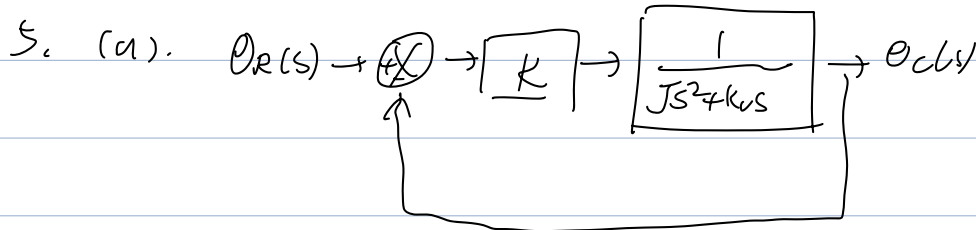
$$\begin{aligned} -9K+2 > 0 &\Rightarrow 9K < 2 \Rightarrow K < \frac{2}{9} \\ K > 0 & \Rightarrow 0, \frac{2}{9} \end{aligned}$$

$$c_2 = -\frac{1}{1/3} \begin{vmatrix} 3 & 0 \\ 1/3 & 0 \end{vmatrix} = 0$$

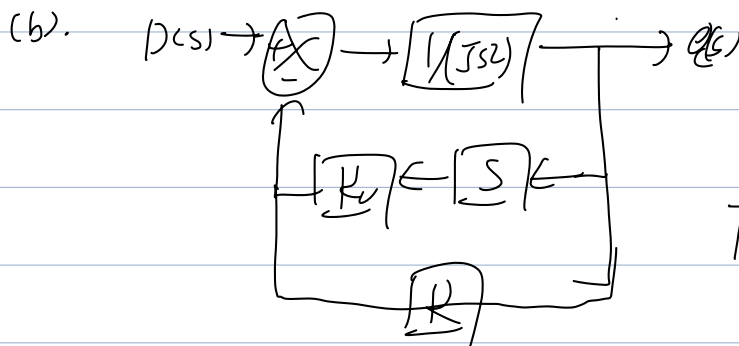
$$\begin{aligned} d_1 &= -\frac{1}{-9K+2} \begin{vmatrix} 1/3 & K \\ -9K+2 & 0 \end{vmatrix} = -\frac{1}{-9K+2} (0 - (-9K+2)K) \\ &= K \end{aligned}$$

1999.

$$\frac{1/(Js^2)}{1 + K_0S/(Js^2)} \quad \frac{\frac{K}{Js^2 + K_0S}}{1 + \frac{K}{Js^2 + K_0S}}$$



$$\Rightarrow \Theta_R(s) \rightarrow \boxed{\frac{K}{Js^2 + K_0S + K}} \rightarrow \Theta_C(s)$$



$$\frac{\frac{1}{Js^2}}{1 + \frac{K_0SK}{Js^2}}$$

$$\Rightarrow D(s) \rightarrow \boxed{\frac{1}{Js^2 + K_0S + K}} \rightarrow \Theta_C(s)$$

$$\begin{aligned} \text{c). } \theta(t) &= \frac{1}{s^2 + ks + k} (K \theta_R(s) + D(s)) \\ &= \frac{1}{25s^2 + 100s + 500} (500 \theta_R(s) + D(s)) \end{aligned}$$

$$\theta_R(s) = \frac{5}{100} \times \pi \cdot \frac{1}{s} = \frac{\pi}{20} \cdot \frac{1}{s}$$

$$D(s) = 50 \cdot \frac{1}{s}$$

$$\theta(s) = \frac{1}{25(s^2 + 4s + 20)} \left(\frac{5\pi}{20} \cdot \frac{1}{s} + 2 \cdot \frac{1}{s} \right)$$

$$= \frac{1}{5(s^2 + 4s + 20)} \cdot 2 \left(\frac{5\pi}{18} + 1 \right)$$

$$= \left(\frac{A}{s} + \frac{Bs+C}{s^2 + 4s + 20} \right) \cdot 2 \left(\frac{5\pi}{18} + 1 \right)$$

$$\Rightarrow \begin{cases} A+B=0 \\ 4A+C=0 \\ 20A=1 \end{cases} \Rightarrow \begin{cases} A = \frac{1}{20} \\ B = -\frac{1}{20} \\ C = -\frac{1}{5} \end{cases}$$

$$\Rightarrow \theta(s) = \left(\frac{1}{20s} - \frac{\frac{1}{20}s + \frac{1}{5}}{s^2 + 4s + 20} \right) \cdot 2 \left(\frac{5\pi}{18} + 1 \right) \cdot \frac{1}{10}$$

$$= \left(\frac{1}{s} - \frac{s+2+\frac{4}{5}}{(s+2)^2 + 4^2} \right) \cdot \frac{1}{10} \left(\frac{5\pi}{18} + 1 \right)$$

$$= \left(\frac{1}{s} - \frac{s+2}{(s+2)^2 + 4^2} - \frac{2}{4(s+2)^2 + 4^2} \right) \cdot \frac{1}{10} \cdot \left(\frac{5\pi}{18} + 1 \right)$$

$$\Rightarrow \theta(t) = \left[1 - e^{-2t} \left(\cos 4t + \frac{1}{2} \sin 4t \right) \right] \cdot \frac{1}{10} \cdot \left(\frac{5\pi}{18} + 1 \right)$$